

## SUMMARY TABLE: PHYSICS GRADE 10

### Sub-Strand 3.3: Introduction to Electronics — Teacher Reference (Pre-filled)

| SUMMARY TABLE: PHYSICS GRADE 10 |                                                                                                                                                                                                                          |
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| Sub-Strand                      | Sub-Strand 3.3: Introduction to Electronics                                                                                                                                                                              |
| Driving Question                | DRIVING QUESTION: Why do some materials conduct electricity while others do not, and how does this affect the safety and efficiency of household wiring and of devices such as solar panels, computers and MRI machines? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                                 | What did I observe?                                                                                                                                                                                                                                                                                                                                                                                                                         | What did I learn?                                                                                                                                                                                                                                                                                                                                                                                      | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                 |
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| <b>Lesson 1</b><br><b>Which Materials Light the Bulb? Launching the Phenomenon</b> | We tested copper wire, glass, dry wood, plastic and a silicon diode in a 3 V circuit with a torch bulb. The bulb was bright with copper, and off with glass, dry wood and plastic. With the diode connected forward (forward-biased) the bulb lit, and with it reversed (reverse-biased) the bulb was off. So 1 of 4 ordinary materials conducts, and the diode conducts in one direction only.                                             | Materials differ in whether they let current flow: copper is a conductor, while glass, wood and plastic are insulators that let almost no current through, which is why Mr Otieno can safely hold plastic-covered copper cable. The diode is different from both groups because it depends on direction.                                                                                               | We cannot yet explain why copper lets current flow and plastic does not, even though both are made of atoms with electrons, or why the diode works one way only. Our first class model shows electrons in each material, but it cannot explain these differences yet. Our DQB questions will guide what we investigate next.                          |
| <b>Lesson 2</b><br><b>Classifying Conductors, Insulators and Semiconductors</b>    | We re-ran the test circuit (3 V, bulb): copper bright; glass, dry wood and plastic off; silicon diode lit forward and off in reverse. The resistivity card showed copper about $1.7 \times 10^{-8}$ ohm metre, germanium 0.46, silicon $2.3 \times 10^3$ , dry wood about $10^8$ to $10^{11}$ , glass about $10^{10}$ to $10^{14}$ and PVC about $10^{13}$ or more. Silicon is about $1.4 \times 10^{11}$ times more resistive than copper. | Materials can be classified by resistivity: conductors (copper) have very low resistivity; semiconductors (germanium and silicon) are in between; insulators (wood, glass, PVC) have very high resistivity and allow almost no current. Conductivity is $1/\rho$ . Household wiring uses copper for efficient current flow and PVC to keep current inside the wire so people can handle cables safely. | We can explain why Mr Otieno uses copper cable covered in PVC for a Nairobi house at 240 V: copper's low resistivity carries current efficiently and PVC's very high resistivity almost stops current reaching a person. We cannot yet explain WHY the resistivities differ so much or why silicon conducts a little, which we will investigate next. |
| <b>Lesson 3</b><br><b>Intrinsic Semiconductors: Silicon and Germanium</b>          | We observed that the silicon diode (1N4007) forward-connected lit the 3 V bulb, as in Lesson 1, and that the data show silicon ( $\rho$ about $2.3 \times 10^3$ ohm metre) and germanium (about 0.46 ohm                                                                                                                                                                                                                                    | We learned that pure (intrinsic) silicon and germanium have a small energy gap (1.1 eV and 0.67 eV), so at room temperature a few electrons gain enough energy to become free electrons, each leaving a hole                                                                                                                                                                                           | We explained that silicon conducts a little because its small gap lets a few electrons cross, unlike glass, wood and PVC with a gap above about 5 eV that allow almost no free electrons, and unlike copper, where the                                                                                                                                |

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|                                                                                          | metre) are far more resistive than copper ( $1.7 \times 10^{-8}$ ohm metre) but far less than glass. The temperature dataset showed the resistivity of pure silicon falling as it was warmed, while copper's resistance rose slightly.                                                                                                                                                                                                                                                  | to make an electron-hole pair. Germanium, with the smaller gap, conducts more than silicon. Warming creates more pairs so conductivity rises.                                                                                                                                                                                                                                                                                                   | bands overlap. This helps explain why semiconductors sit between copper wire and its plastic cover in Mr Otieno's Nairobi house, and raises the question of how to control their conduction, which we will investigate next.                                                                                                                                                                                                                                         |
| <b>Lesson 4</b><br><b>Doping: Making n-type and p-type Semiconductors</b>                | We re-tested the 1N4007 diode: the bulb lit when it was forward-connected (forward-biased) and stayed off when it was reverse-connected (reverse-biased). Using counters, we saw that replacing one silicon atom with phosphorus (5 outer electrons) leaves an extra free electron, and replacing one with boron (3 outer electrons) leaves a hole. The data card showed that doping about 1 atom in $10^7$ to $10^8$ raises the conductivity of silicon by a factor of many thousands. | Doping makes extrinsic semiconductors from pure (intrinsic) silicon. Phosphorus gives n-type (majority carriers: electrons) and boron gives p-type (majority carriers: holes). Doped silicon is still electrically neutral overall. Doping lets us control conductivity, and joining p-type and n-type makes a p-n junction, which is the diode.                                                                                                | Pure silicon conducts very poorly (resistivity about $2.3 \times 10^3$ ohm metre), but a tiny amount of phosphorus or boron adds free electrons or holes, so conductivity rises thousands of times in a controlled way. A p-n junction lets current pass when forward-biased (bulb lit) and blocks it when reverse-biased (bulb off), which is why doped silicon is the basis of diodes, solar panels like those at Garissa, and microchips in phones and computers. |
| <b>Lesson 5</b><br><b>PhET Simulation: Testing the Band Model with Digital Evidence</b>  | What did the simulation show when we changed the energy gap (metal 0 eV, germanium 0.67 eV, silicon 1.1 eV, insulator above about 5 eV), raised the temperature of silicon, and added phosphorus or boron? Record the numbers and the number of free electrons and holes you saw.                                                                                                                                                                                                       | What did the simulation evidence tell us about how well our band model works? Which parts did it support, which parts did we have to change, and what are the limits of a simulation compared with the real bulb test in the Nairobi wiring circuit?                                                                                                                                                                                            | Using the evidence, explain why copper lights the bulb while glass, wood and plastic almost do not, why silicon sits in between and conducts more when warmer, and how phosphorus (n-type) and boron (p-type) change it. Link your answer to why Mr Otieno can safely hold PVC-covered copper cable.                                                                                                                                                                 |
| <b>Lesson 6</b><br><b>Superconductors and MRI Magnets: When Resistance Drops to Zero</b> | We observed that a YBCO disc at room temperature did not levitate a magnet, but cooled to 77 K (-196 C) in liquid nitrogen it made the magnet float. We compared Tc values: mercury 4.2 K, NbTi 9.2 K, YBCO about 92 K, and noted which coolant (77 K or 4.2 K) is cold enough for each.                                                                                                                                                                                                | We learned that a superconductor has zero resistance below its critical temperature ( $T_c$ ), unlike copper, which always has some resistance ( $1.7 \times 10^{-8}$ ohm metre at about 20 C). With zero resistance a large current flows with no energy lost as heat.                                                                                                                                                                         | We explain that Dr Wekesa's MRI machine uses NbTi coils cooled to 4.2 K with liquid helium, below the $T_c$ of 9.2 K, so the magnet carries a large current without energy loss and gives a field of about 1.5 T to 3 T. Household wiring still uses copper and PVC because room temperature is far above every $T_c$ we studied.                                                                                                                                    |
| <b>Lesson 7</b><br><b>Case Studies and the Final Answer to the Driving Question</b>      | What we observed: copper lit the bulb brightly; glass, dry wood and plastic did not; the silicon diode (1N4007) lit the bulb only when forward-connected. In the Kenyan cases we saw silicon p-n junctions at work in the Garissa solar plant (50 MW) and in microchips, and an NbTi superconducting magnet cooled to 4.2 K in the Kenyatta National Hospital MRI, with a YBCO disc levitating a magnet at 77 K.                                                                        | What we learned: materials differ in resistivity (copper $1.7 \times 10^{-8}$ ohm metre up to PVC about $10^{13}$ or more) and in energy gap (metals 0 eV, germanium 0.67 eV, silicon 1.1 eV, insulators above about 5 eV). Doping silicon with phosphorus (n-type) or boron (p-type) raises conductivity in a controlled way, and a p-n junction conducts one way only. Superconductors have zero resistance below their critical temperature. | How we explained it: Copper conducts because its outer electrons are free to move (bands overlap), while glass, wood and plastic have a large gap so almost no electrons can move, making PVC a safe cover for Mr Otieno's wires. Silicon and germanium conduct a little, more when warmer, and doping with phosphorus or boron greatly raises conductivity, so p-n junctions make solar panels and microchips                                                       |

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|  |  |  | work. Below $T_c$ , NbTi has zero resistance so MRI magnets carry large current without energy loss, which answers our Driving Question. |
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**ATTRIBUTION AND LICENSING**

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